

Timbol Steven B. Geometry
ICT 106

Problem Set 7: Distance Between Two Points and Division of a line segment.

1. Find the midpoint of (AB) whose end points are A(7,3) and B(5,-1)

$$m = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \quad m = \left(\frac{7+5}{2}, \frac{3+(-1)}{2} \right) \quad m = \left(\frac{12}{2}, \frac{2}{2} \right) = (6, 1)$$

2. The midpoint of a line segment is (-1,7). If one end point is (3,4) find the other endpoint.

$$m = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \quad (-1, 7) = \left(\frac{3 + x_2}{2}, \frac{4 + y_2}{2} \right)$$
$$\begin{aligned} -1 &= \left(\frac{3 + x_2}{2} \right) \rightarrow -2 = 3 + x_2 \rightarrow x_2 = -5 \\ 7 &= \left(\frac{4 + y_2}{2} \right) \rightarrow 14 = 4 + y_2 \rightarrow y_2 = 10 \end{aligned}$$

end point = (-5, 10)

3. Show that the points A(6,2) B(2,1) C(1,5) and D(5,6) are the vertices of a square.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$1. AB = \sqrt{(2-6)^2 + (1-2)^2} = \sqrt{(-4)^2 + (-1)^2} = \sqrt{16+1} = \sqrt{17}$$

$$2. BC = \sqrt{(1-2)^2 + (5-1)^2} = \sqrt{(-1)^2 + (4)^2} = \sqrt{1+16} = \sqrt{17}$$

$$3. CD = \sqrt{(5-1)^2 + (6-5)^2} = \sqrt{(4)^2 + (1)^2} = \sqrt{16+1} = \sqrt{17}$$

$$4. DA = \sqrt{(6-5)^2 + (2-6)^2} = \sqrt{(1)^2 + (-4)^2} = \sqrt{1+16} = \sqrt{17}$$

Diagonals:

$$1. AC = \sqrt{(1-6)^2 + (5-2)^2} = \sqrt{(-5)^2 + (3)^2} = \sqrt{25+9} = \sqrt{34}$$

$$2. BD = \sqrt{(5-2)^2 + (6-1)^2} = \sqrt{(3)^2 + (5)^2} = \sqrt{9+25} = \sqrt{34}$$

$$\sqrt{17} \quad \& \quad \sqrt{34}$$

4. The point P divides the line segment ~~AB~~^{BA} joining points A(6, 2) and B(-4, -6) in the ratio 3:1. Find the Point of Division.

$$P = \left(\frac{r_1 x_2 + r_2 x_1}{r_1 + r_2}, \frac{r_1 y_2 + r_2 y_1}{r_1 + r_2} \right)$$

$$P = \left(\frac{3(-4) + 1(6)}{3+1}, \frac{3(-6) + 1(2)}{3+1} \right)$$

$$P = \left(\frac{-12+6}{4}, \frac{-18+2}{4} \right) = \left(\frac{-6}{4}, \frac{-16}{4} \right) = (-1.5, -4)$$